Printed Page: 8 **SET A** University Roll No:.....

End-Term Examination, Odd Semester: 2021-22

B.Tech (CSE/CCV/DA/CSF/AIML), II Year, Semester III

BELH-0006: Ethics & Values

Time: 2.00 hrs Max. Marks: 50

All questions carry equal marks: **0.5 X 100 = 50**

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a. Not true b. False c. legal case d. none of these.
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a. Three versions b. Results are same in three versions c. both a&b d. None of these.
3.Trolley problem primarily involves:
a. Utilitarianism b. Deontology c. Double effects d. all above.
4. The majority decision in the first version is not :
a. Change track b. Go straight c. stop the trolley d. save the five
5. The majority decision in the second version:
a. Do not push the fat man b. change the track c. save the five d. none of these.
6. Majority don't push the fat man because:
a. Right action b. Fat man is heavy c. Wrong intention d. Maximum Good
7. Majority change the track because:
a. On Deontology b. On Virtue c.On Utilitarianism d. both a & c
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a. Only EDM b. Only Utilitarianism c. Deontology d. None of these
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a. Phillipa Foot b. J.S Mill c. James d. Bentham
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a. No influence of technology b. No Utilitarianism c. both a&b d. None of these.
11. The case of cannibalism is:
a. Hypothetical case b. unreal case c. real case d. all of these
12. Who was the captain:
a. Richard Parker b. Dudley c. Stephenson d. Brooke

13. Who was the killer of Richard Parker:

a. Brooke b. Stephenson c. Richard Parker d. None of these

14. Parker was:

a. Orphan b. youngest c. ill d. all above

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a. Capacity to act b. capacity to think c. capacity to cheat d. none of these

22. Story of stuff is about:

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a. Non-Linear b. Non-recyclable c. Unsustainable d. all of these

24. What is green production:

a. Environment Unfriendly b. Producing carbon c. Harming nature d. None of these

25. Market production advertises that:

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1. My personality is effective in my being free to choose anything.
2. I am respected person when everyone follows me.
3. My decisions are good when no one dare to question.
4. My personality is good when I am able to satisfy my desires.
5. I am rational only in fulfilling my interests.
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96. "I am right only in being good and am good only in being right". What are the reasonable objections to this statement?

1. This is not a valid statement for all the cases.
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4. A right always results in good.
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97. "In order to be a good professional: We should be focused on our value judgments". What we can infer from this statement?
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5. The majority decision in the second version:
a. Do not push the fat man b. change the track c. save the five d. none of these.
6. Majority don't push the fat man because:
a. Right action b. Fat man is heavy c. Wrong intention d. Maximum Good
7. Majority change the track because:
a. On Deontology b. On Virtue c.On Utilitarianism d. both a & c
8. Trolley problem involves:
a. Only EDM b. Only Utilitarianism c. Deontology d. None of these
9. Who designed the trolley problem:
a. Phillipa Foot b. J.S Mill c. James d. Bentham
10. Third version shows:
a. No influence of technology b. No Utilitarianism c. both a&b d. None of these.
11. The case of cannibalism is :
a. Hypothetical case b. unreal case c. real case d. all of these
12. Who was the captain :
a. Richard Parker b. Dudley c. Stephenson d. Brooke

13. Parker was:
a. Orphan b. youngest c. ill d. all above
14. Who was the killer of Richard Parker:
a. Brooke b. Stephenson c. Richard Parker d. None of these
15. Stephenson intended killing Parker:
a. Yes b. No c. not sure d. none of these.
16. Argument from necessity fails in the view of :
a. Certainty b. Uncertainty c. saved Dudley d. Justified Killing Parker
17. What was the utilitarian argument in this case:
a. From Necessity b. From Intention c. From moral agency d. saving more lives
18. No lottery drawn in the case but If lottery would have been drawn then it
a.Fully justify the case b. Partially justify the case c. not sure d. all of these
19. In this case of cannibalism:
a. Dudley exonerated b. Brooke killed c. given capital punishment d. none of these
20. Killing Parker was:
a. Deontologically right b. ethically right c. ethically wrong d. both a&b.
21. Story of stuff shows:
a. Capacity to act b. capacity to think c. capacity to cheat d. none of these
22. Story of stuff is about:
a. Personal ethics b environmental ethics c. both a & b d. only a
23. What is green production:
a. Environment Unfriendly b. Producing carbon c. Harming nature d. None of these
24. Production system is wrongly directed because it works:
a. Non-Linear b. Non-recyclable c. Unsustainable d. all of these
25. Market production advertises that:
a. You are not fit b. Update with new c. both a&b d. none of these
26. Modern times gives a message:
a .Keep trying b. Keep Idle c. Be happy d. Both a&c
27. What is depicted in the first scene of the documentary: Modern Times?
a. Mechanical system b. Ethical System c. Inhuman System d. None of these
28. The girl in the movie was a real thief :
a. Yes b. No c. Not sure d. Uncertain

29. The Modern Times is a criticism of :

a. Only Mechanical World b. Unethical world c. Only Ethical World d. None of these

30. Jailer was happy with Charlie because:

a.He helped system b. He helped for himself c. Only A d. All above

31. Right Here Right Now shows:

a. Personal ethics matters b. professional ethics only c. only happiness d. none of these

32. Dr was right in not meeting the girl:

a.Yes b. No c. not sure d. professional ethics was first

33. Driver rents bicycle for boy because :

a. Shared happiness b. had spare money c. had nothing to buy d. none of these

34. Elder brother praised sketching did not lead:

a. Negative b. maid left the job c. accident d. all of these

35. Old women said daughter to marry south Indian:

a. She understood human values b. professional ethics c. self interest d. all of these

36. Who worked on the Utilitarianism theory:

a. Jeremy Bentham b. J.S. Mill c. both d. non one

37. According to Bentham what matters:

a. Pleasure b. quality c. quantity d. both a & c

38. Who developed pleasure calculus:

a. Kant b. Mill c. Bentham d. John Rawls

39. Higher pleasure, being human and anti slavery arguments are discussed in the book:

a. On Liberty b. Utilitarianism c. Liberty d. none of these

40. Who wrote Utilitarianism:

a. I. Kant b. Mill c. James d. none of these

41. Quarantine practice is justified on the principle of :

a. Utility b. social welfare c. maximum good d. all of these

42. Who defended psychological hedonism for ethical theory:

a. Mill b. Bentham c. Kant d. none of these

43. Mill gives priority to liberty over

a. Greater good b. Pleasure c. not sure d. both a & b

44. Human being are different from chimps because they can make:
    a. Quality difference b. choice c. analyze d. all of these

45. Choice making involves:
    a. Rationality b. quality judgments c. difference d. all of these
46. Libertarianism is based on:
    a. Individual rights b. Greater Good c. Pleasure d. all of these
47. Liberty according to libertarians:
    a. All or nothing b. nothing c. Complete d. a&c
48. The purpose of morality is to maintain individual rights:
    a. Liberty only b. Respect c. Ownership only d. none of these
49. Libertarians argue for minimal state theory because
    a. To maintain liberty b. state excessively intervene c. both d. both not
50. Respect means:
    a. Treating individual as ends b. As means c. As humans d. For profit
51. What are the objections by Libertarians:
    a. Taxation b. no moral legislation c. paternalistic laws d. all of these
52. Libertarians objects Moral Legislation because:
    a. State should give moral code b. State should develop values c. both d. both not
53. Paternalistic laws are like
    a. Driving safety rule b. pension rules c. both not d. a & b
54. Taxation is wrong as it makes once profession:
    a. Slavery b. forced labor c. shares ownership d. all of these
55. Paternalistic laws are against liberty because
    a. They undermine risk b. freedom only c. respect only d. all of these
56. Sate should not protect individuals from
    a. Oneself b. not from external threats c. not from cheat d. all of these
57. Robert Nozick rejected :
    a. Taxation on salary b. Taxation c. Taxation on Individual Salary d. none of these
58. Libertarianism and Utilitarianism are different as on
    a. Purpose of morality b. individual rights c. both a& b d. both not
59. What is affirmative action?
    a. For reservation only b. For correcting past c. for social justice only d. None of these

60. Corrective argument put emphasis on
    a. Correcting past justice b. correcting future c. against reservation d. none of these

61. Compensatory argument put emphasis on
    a. No Reservation as compensation b.reservation c.for past wrong d. both b & c
62. Social representation argument put emphasis on :
    a. Maintain variety of groups b. maintain reservation c. Both d. Both not
63. Reservation has not achieved is purpose because
    a. Reached to some deserving b. reached undeserving c. both d. both not
64. Reservation may not be right if given
    a. For poor economic sections b. For basic resources c.for job d.None of these
65. Morality is good in itself according to
    a. Deontology b. Utilitarianism c. Virtue d. Libertarianism
66. What is supreme principle of morality?
    a. Categorical imperative b. hypothetical imperative c. both d. both not
67. Imperative means:
    a. Moral command b. legal command c. emotional command d. political command
68. Hypothetical imperative is
a. Not based on ends b. unconditional c. Not based on result d. all of these
69. Humanity principle emphasizes
a. Humanity as an end b. Humanity as means c. Humanity as need d. Humanity as greed
70. According to Kant what matter is
a. Motive b. results c. Intentions d. both a & c
71. A maxim can be a moral law iff and only:
    a. Universalizable b. happiness c. goodness d. all of these
72. The Ground works of the metaphysics of morals includes the ideas:
    a. Autonomy b. Rationality c. Moral law d. all of these
73. According to John Rawls what matter is
    a. Only Fairness b. unfairness c. fair distribution d. none of these
74. The difference between a maxim and a moral law underlies
    a.Objectivity b. Universality c. Individuality d. only a & c

75. Ideas discussed in the book A theory of Justice include
    a. Well ordered society b. Cooperative Society c. Both d. Both Not

76. Rawls give two principles of justice:
    a. Principle of freedom b. Principle of unqualified freedom c. both d. both not
77. Veil of ignorance is a thought experiment that is for making
    a. Equality b. removing natural imbalances c. removing natural inequality d. all of these
78. According to Rawls justice means
    a. Fair distribution b.distribution in basic resources c.both a & b d. none of these
79. Rawls argues justice is
    a. Correcting wrong b. correcting natural injustice c. both d. both not
80. Rawls rejects
    a. Only Utilitarianism b. only Feudalism c. Meritocracy d. none of these
81. India allows which kind of Euthanasia :
    a. Active b. Passive c. both d. none of these
82. What is legitimate entitlement?
    a. What you deserve b. legitimately deserve c. what is entitled d. both b & c
83. Justice is teleological & attributive:
    a. Aristotle b. Rawls c. Kant d. Mill
84. Euthanasia refers to
    a. Mercy killing b. Killing on once choice c. both a & b d. only a

Statement Based Questions

85. "Each virtue has a purpose what Aristotle means by this statement".
    a. Virtue without practice b. merits c. Virtue with practice d. all of these
86. "You deserve on what you do". What this statement means to:
    a. Moral Desert b. Moral freedom c. Justice d. Moral decision
87. "Best flute players deserve the bast flutes". What does it mean:
    a. Only Best merit b. best practices c. best utility d. both a& b
88. India bans which kind of surrogacy?
    a. Philanthropic b. Commercial c. both a & b d. all of these
89. "A woman has to have the right for surrogacy".
    a. No b. Yes c. In some cases d. both b& c
90. "The legal right to life also includes right to respectful death".
    a. Moral right b. legal right c. human right d. All of these
91. "One owns the life so euthanasia should be morally permissible".

a. Libertarians b. Active euthanasia c. Passive Euthanasia d. all of these

92. " Life is not a joy machine to have happiness all the time".

a. Kantian b. Libertarian c. utilitarian d. all of these

93. "Choice is a representation of my rationality and I am free to do whatever I desire to choose". What is wrong in the statement?

1. Choice is not the freedom to choose anything.
2. Choosing anything does not represent my rationality.
3. Freedom is not defined by my desires.
4. I am not free to choose anything.
5. I am free to choose reasonable course of actions.
6. All above.

a. only 1& 2 b. Only 2 & 3 c. only 6 d. 1,2,3,4

94. "I am a rational person in my practical affairs when I am scientific and objective in my point of views". What does logically imply from the statement?

1. I cannot have my individual point of view.
2. I can have only my subjective point of view.
3. I can have reasonable point of views.
4. I can have the reasons suitable to self interests only.
5. I have to mind only what I find right.
6. All above.

a. only 6 b. only 1,2 & 3 c. only 3 d. Only 5

95. "Choices, decisions and actions create our personality". What has been irrationally inferred in the following statement?

1. My personality is effective in my being free to choose anything.
2. I am respected person when everyone follows me.
3. My decisions are good when no one dare to question.
4. My personality is good when I am able to satisfy my desires.
5. I am rational only in fulfilling my interests.
6. All above.

a. only 1,2 ,3 b. Only 6 c. Only 2, 3, 4 d. only 3, 4, 5

96. "I am right only in being good and am good only in being right". What are the reasonable objections to this statement?

1. This is not a valid statement for all the cases.
2. This is a valid statement for all the cases.
3. Sometimes a right results in bad.
4. A right always results in good.
5. Sometimes the right actions produce the good results.
6. None of the above.

a. 1, 3 & 5 b. Only 6 c. only 1, 2, 3 d. only 3, 4 & 5

97. "In order to be a good professional: We should be focused on our value judgments". What we can infer from this statement?
1. Long term goals should be given priority.
2. Short term profits are must.
3. Short term good can be sacrificed for a longer term good.
4. Long term reputation matters.
5. Short term pleasure matters.
6. Only profit matters.
7. All above.
a. only 7 b. only 4,5 & 6 c. only 2,3 & 4 d. 1, 3 & 4

98. "The intention of only job for engineering is not right"
a. Teleopathy b. Teleology c. Tautology d. Technology
99. "Your life is not like a falling object or a billiards ball".
a. Humans beyond the gravity b. Moral freedom c. both d. both not

100. "Not we have the winners but losers in the nature".
a. Natural justice b. attributive justice c. both d. both not

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End Term Examination, Odd Semester 2021-22

B. Tech , CSE, Year: II , Semester: III

Subject Code: BCSC0008 **Subject Title Computer Networks**

Time: 3 Hour **Maximum Marks: 50**

Section- A

Note: Attempt ANY FIVE Questions. 5 x 4 = 20 marks

(I) a) Suppose two friends have created a network using Bluetooth. Mention the name of IEEE standard that these friends will be making use of.

b) What is the relationship between period and frequency?

(II) a) What is vulnerable time in slotted aloha?

b) How do guided media differ from unguided media?

(III) a) List the name of protocol or protocols, in which it is possible for the sender to receive on ACK for a packet that falls outside its current window?

b) 50 employees are working in an organization and everybody want to link with everyone. If the length of one link is 5 meters, then how much length of cable required?

(IV) a) How many bits are allocated for NID and HID IN 23.197.157.234?

b) A host with IP address 200.100.1.1want to send a packet to all host in same network, what is the source IP address and destination IP address

(V) a) Compare and contrast a circuit-switched network and a packet- switched network.

b) What is difference between bit rate and baud rate? Explain it with suitable diagram?

(VI) a) An IPv4 packet has arrived with the first 8 bits as shown: 01000010 The receiver discards the packet. Why?

b) Explain the role of Network Layer?

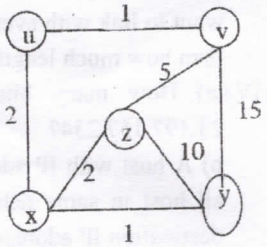
Section- B

Note: Attempt ALL Questions. 5 x 3 = 15 marks

(I) A block of addresses is given to a small organization. One of the addresses is 215.16. 38.39/28, What is the

i. First address
ii. Last address
iii. No. of addresses

(II) Consider the network shown below, and assume that each node initially knows the costs to each of its neighbors, Consider the distance vector algorithm and show the distance table entries at node z



(III) Discuss the three-way Handshaking process of TCP connection management.

(IV) Draw and Explain IPv4 Datagram Header?

(V) What do you understand by cryptography? explain the public and private key cryptography?

Section- C

Note: Attempt ANY THREE Questions. 3 x 5 = 15 marks

(I) Consider an instance of TCP's Additive Increase Multiplicative Decrease(AIMD) algorithm where the window size at the start of the slow start phase is 2 MSS and the threshold at the start of the first transmission is 8 MSS. Assume that a time out occurs during the fifth transmission. Find the congestion window size at the end of the tenth transmission

(II) If Classless Inter-domain Routing (CIDR) receives a packet with address 180.70.65.140. The router's routing table has the following entries:

Mask	*Network Address*	*Next Hop*	*Interface*
/26	180.70.65.192	—	m2
/25	180.70.65.128	—	m0
/24	201.4.22.0	—	m3
/22	201.4.16.0	—	m1
Any	Any	180.70.65.200	m2

Identify output interface on which this packet will be forwarded.

(III) Explain the difference between IPv4 and IPv6?

(IV) Suppose a network with IP Address 192.16.0.0. is divided into 2 subnets, find number of hosts per subnet.
Also, for the first subnet, find-

i. Subnet Address
ii. First Host ID
iii. Last Host ID
iv. Broadcast Address

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End Term Examination, Odd Semester 2021-22
B. Tech., CS (Sec. E, F, G, H, I, J, K, L), EE, EN, EC and CE,
II Year, III Semester
BMAS 1103: Engineering Mathematics III

Time: 3 Hours **Maximum Marks: 50**

Section- A

Note: Attempt ANY FIVE Questions. 5 x 4 = 20 marks

Q.1. Comment on the following statements:

(a) For a binomial distribution, mean is 6 and variance is 9.

(b) For a Poisson distribution with number of trials n = 1000 and probability of happening of an event p = 0.002, mean and variance are 2.

Q.2. The regression lines of y on x and x on y are respectively,

$y = a x + b$,

$x = c y + d$. (a, b, c and d are constants)

Can you determine the mean values of x and y? If yes, find them.

Q.3. Evaluate: $L^{-1}[\frac{p+8}{p^2+4p+5}]$

Q.4. What are Cauchy- Riemann equations in polar coordinates? Show that $f(z) = r^2(\cos 2\theta + i \sin 2\theta)$ is analytic.

Q.5. What is Cauchy's integral theorem? Evaluate:

$$\oint_c \frac{e^{-z}}{z+1} dz$$

where c is the circle with centre at z = 0 and radius 0.5.

Q.6. What is a singularity? Discuss the nature of singularity of

$$f(z) = \frac{z - \sin z}{z^3} \quad \text{at } z = 0.$$

Section- B

Note: Attempt ALL Questions. 5 x 3 = 15 marks

Q.1. Examine the nature of the function

$$f(z) = \frac{x^2y^5(x+iy)}{x^4+y^{10}} ; z \neq 0 \text{ and } f(0) = 0$$

in the region including the origin.

Q.2. Determine the constants a, b, c and d so that the function

$$f(z) = (x^2 + a\,x\,y + by^2) + i\,(c\,x^2 + d\,x\,y + y^2)$$

is analytic.

Q.3. Expand the function $f(z) = \frac{1}{z-2} - \frac{1}{z-1}$ in the regions:

(a) $|z| < 1$

(b) $1 < |z| < 2$

(c) $|z| > 2$.

Q. 4. Show that:

$$\oint_c \frac{e^z}{z}\, dz = 2\pi i \quad \text{where c is } |z| = 1.$$

Hence show that,

$$\int_0^{2\pi} e^{\cos\theta} \cos(\sin\theta)\, d\theta = 2\pi$$

Q. 5. Find the pole of the function

$$f(z) = \frac{1 - e^{2z}}{z^4}$$ and determine its order.

Find also the residue of $f(z)$ at that pole.

Section- C

Note: Attempt ANY THREE Questions. 3 x 5 = 15 marks

Q.1. Using contour integration, evaluate:

$$\int_0^{2\pi} \frac{d\theta}{a + b\cos\theta} \; ; a > |b|$$

Q.2. If $u - v = (x - y)(x^2 + 4xy + y^2)$ and $f(z) = u + iv$ is an analytic function of $z = x + iy$ then find $f(z)$ in terms of z.

Q.3. What is the bilinear transformation which maps the points $z = \infty, i, 0$ of z – plane into the points $w = 0, i, \infty$ of w- plane?

Q.4. Integrate $f(z) = Re\,(z)$ from $z = 0$ to $z = 1 + 2i$,

(a) Along the straight line joining $z = 0$ and $z = 1 + 2i$.

(b) Along the real axis from $z = 0$ to $z = 1$ and then along a line parallel to imaginary axis from $z = 1$ to $z = 1 + 2i$.

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End-Term Examination, Odd Semester 2021-22

B.Tech (CSE/CCV/DA/IoT/AIML/CSF)

IInd Year, IIIrd Semester

BCSC1002: Object-Oriented Programming

Time: 3 Hour **Maximum Marks: 50**

Section-A

Note: Attempt ANY FIVE Questions. 5 x 4 = 20 marks

Q1. Create a base class with an abstract print( ) method that is overridden in a derived class. The overridden version of the method prints the value of an int variable defined in the derived class. At the point of definition of this variable, give it a nonzero value. In the base-class constructor, call this method. In main method, create an object of the derived type, and then call its print( ) method. Explain the results.

Q2.

a) Write a java statement for each to perform the following task:

a. Convert a number stored in a string variable str to double data type.

b. Extract the second character of the string str.

b) Consider the following code:

```
Class.forName("mypackage.MyClass");
Connection con = DriverManager.getConnection("Some connection string");
```

Which exception can be thrown in the above statements?

Q3. Design a class to overload a function Sum() as follows:

I) int Sum(int A, int B) – with two integer arguments (A and B) calculate and return sum of all the even numbers in the range of A and B.
Sample input: A=3 and B=10 Sample output: sum = 4 + 6 + 8 + 10

II) int Sum(int N) - with one integer argument (N) calculate and return sum of only odd digits of the number N.
Sample input: N=43961 Sample output: sum = 3 + 9 + 1 = 13

Write the main method to create an object and invoke the above methods.

Q4. Explain two situations when String or StringBuffer class would be used for string handling. Also write a program which appends the string "programming", to string "java". Print the final content of the appended string.

Q5. Write a program with division method who receives two integer numbers and performs the division operation. The method should declare that it throws ArithmeticException. This exception should be handled in the main method.

Q6.

a) The following code fragment contains error. Can you point it out?

```
class Test
{       int demon = 3;
        static int value = 1024;
        static int valueDivdenom()
        {
                return value/denom;
        }
}
```

b) what will be the output of the below program? Also give explanation.

```
class A
{ void m1() throws IOException
  {
    System.out.println("In m1 A");
  }
}
class B extends A
{ void m1() throws Exception
  {
    System.out.println("In m1 B");
  }
}
```

```
public class Test {

  public static void main(String[] a)
{
    A a=new B();
    try {
      a.m1();
    } catch (IOException e) {
      e.printStackTrace();
    }
  }
}
```

Section-B

Note: Attempt ALL Questions. 5 x 3 = 15 marks

Q1. Using Regular Expression syntax for Java, create a program to represent all valid identifier.
The rules are:-

a) The allowed characters are a-z A-D 0-9 $
b) The length of the identifier should be at least 3
c) The first character should be lowercase alphabet symbol from a to m
d) The second character should be digit divisible by 2 (2,4,6,8)

Q2. Why do we need Generics? Can you give an example of how Generics make a program more flexible?

Q3. Explain the hierarchy of collection framework in java. Differentiate between List and Set in Java.

Q4. What is the difference between Statement and PreparedStatement. Write the statement to call a stored procedure from java language by using CallableStatement.

Q5. Write a program in Java to display the contents of a LinkedList using Iterator.

Section-C

Note: Attempt ANY THREE Questions. 3 x 5 = 15 marks

Q1. Consider a class that represents an account in a bank. Let the minimum balance for the account be rupees 1000. WAP that throws a custom exception when a withdrawal request results in balance decreasing from minimum balance.

Q2. Write a java program to setup JDBC and execute an SQL statement on a database table "BOOK", with the fields BOOK_ISBN, BOOK_AUTH, BOOK_PRICE, BOOK_PUBLISH. The SQL statement should find the BOOK_AUTH and BOOK_PRICE of the book having ISBN= 1236.

Q3. What is a thread? How is it different from a process? Explain the basic constructs to create threads and use them to synchronize asynchronous tasks by creating program.

Q4. Create a collection to store objects of product class by using ArrayList. The product class would contain the following three private fields:-

String productname, int productid, double productprice

print the product details according to the productprice in descending order.

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End Term Examination, Odd Semester 2021-22

B. Tech, CSE, II- year, III-Semester

Database Management Systems BCSC 0003

Time: 3 Hour **Maximum Marks: 50**

Section A

Note: Attempt ANY FIVE Questions **(5x4=20 marks)**

I. (a) What do you understand by Entity integrity constraints explain it with suitable example?

(b)Differentiate between database schema and database state?

II. Differentiate between single valued attribute vs multiple valued attributes with suitable diagram.

III. (a) Define the term redundancy and consistency.

(b) Why normalization is needed.

IV. (a) What do you understand by serial and non-serial schedule?

(b) Explain the incorrect summary problem with suitable example?

V. Discuss the different type of failure in transactions.

VI. What do you understand distributed database?

Section –B

Note: Attempt ALL Questions. **(5x3=15 marks)**

I What do you understand by transaction processing system?

II Explain the desirable properties of Transaction?

III What do you understand by data fragmentation and also explain all fragmentation?

IV Is the following schedule conflict equivalent? Justify.

S1:R1(A)W1(A)R2(B)W2(B)R1(B)

S2:R1(A)W1(A)R1(B)R2(B)W2(B)

V. Is the following schedule conflict serializable?
S1:R4(A)R2(A)R3(A)W1(B)W2(A)R3(A)W2(B)

Section- C

Note: Attempt ANY THREE Questions. 3x5 = 15 marks

(I) What do you understand by two phase locking protocol? Is the following 2PL, strict, rigorous, conservative?

L_S(A)
R(A)
L_X(B)
R(B)
U(A)
W(B)

(II) What is schedule? Define the concept of recoverable, cascade less and strict schedules, and compare them in term of their recoverability.

(III) What is the limitation in locking based techniques so that timestamp-based techniques are required?

(IV) Let a relation R (A, B, C, D) and functional dependency {AB –> C, C –> D, D –> A}. Relation R is decomposed into R1(A, B, C) and R2(C, D). Check whether decomposition lossless and dependency preserving or not.

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End Term Examination, Odd Semester 2021-22

BTech (CSE) - II Year / III Semester

Operating System (BCSC0004)

Time: 3 Hour **Maximum Marks: 50**

Section- A

Note: Attempt any FIVE Questions. **5 x 4 = 20 marks**

1. What is Semaphore? Explain in detail the Reader & Writer Problem of process synchronization with suitable code (both Reader and Writer Process).
2. Explain Pre-emptive and Non Pre-emptive CPU scheduling algorithm? Describe Multilevel feedback queue scheduling in brief!
3. Describe Inter Process Communication (IPC) and their method to implement IPC in the system?
4. How deadlock condition occurs? What are the necessary and sufficient condition for a system to be in deadlock? Explain in detail!
5. The Logical memory and physical memory is 32k and 512 words respectively. If Page size is of 64 words, formulate logical and physical memory space. How many pages logical and physical memory can accommodate? Write number of entries in the page table also.
6. Write the possible protocols to prevent "**Hold and Wait**" and "**Circular Wait**" condition for Deadlock Prevention? Clearly Explain!

Section- B

Note: Attempt ALL Questions. **5 x 3 = 15 marks**

1. What is external fragmentation? What are the probable solution for the removal of external fragmentation? Explain!
2. Considering a system with five processes P_0 through P_4 and three resources types A, B, C. Resource type A has 10 instances, B has 5 instances and type C has 7 instances. Suppose at time t_0 following snapshot of the system has been taken:

Process	Maximum Need			Allocation		
	A	B	C	A	B	C
P0	7	5	3	0	1	0
P1	3	2	2	2	0	0
P2	9	0	2	3	0	2
P3	2	2	2	2	1	1
P4	4	3	3	0	0	2

a. Find the Need Matrix?
b. Find whether system is in Deadlock or not? Write the safe sate (if Possible)
c. What will happen if process P_1 requests one additional instance of resource type A and two instances of resource type C? Is additional request by process P1 granted?

3. A virtual memory system has an address space of 16K words, a memory space of 6K words, and page and block sizes of 2K words The following page reference changes occur during given time interval.
   2 4 0 1 2 6 1 4 0 1 0 2 3 5 7
   Determine the three pages that are resident in main memory after each page reference change if the replacement algorithm used is (a) FIFO; (b) LRU.
4. Given five memory partitions of 500 KB, 350 KB, 250 KB, 420 KB, and 450 KB (in order), how would the, best fit and worst-fit algorithms place processes of 325 KB, 150 KB, 400 KB, and 375 KB (in order)?
5. Write short notes on the following
   a. Demand Paging
   b. Thrashing

Section- C

Note: Attempt any THREE Questions. **3 x 5 = 15 marks**

1. What is file System? Explain various of directory structure with suitable example and diagram.

2. What is safe sequence? Write and explain Banker's Algorithm for Dead avoidance.
3. Draw the resource allocation graph and determine deadlock or deadlock free state.

$P = \{P_1, P_2, P_3\}$
$R = \{R_1, R_2, R_3, R_4\}$
$E = \{ P_1 \rightarrow R1, P2 \rightarrow R3, R1 \rightarrow P2, R2 \rightarrow P2, R2 \rightarrow P1, R3 \rightarrow P3, P3 \rightarrow R2\}$
There are ONE, TWO, ONE and THREE instances of R_1, R_2, R_3 and R_4 type resources

4. In a paged-segmented system, a virtual address consists of 32 bits of which 12 bits are displacement, 11 bits segment number and 9 bits are page numbers.
   Calculate the following:
   a. Page size
   b. Max segment size
   c. Max number of Pages
   d. Max number of segments

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End Term Examination, Odd Semester 2021-22

B.Tech. CSE/AIML/CSF/CCV/DA/IIoT, II Year, III Semester

BCSC1005: Computer Organization

Time: 3 Hour **Maximum Marks: 50**

Section- A

Note: Attempt ANY FIVE Questions. 5 x 4 = 20 marks

Q1. The 8 bit registers AR, BR, CR, DR initially have the following values.
AR = 11110010 BR = 11111111 CR = 10111001 DR = 11101010

Determine the final 8-bit values in each register after the execution of the following sequence of micro-operations.
AR=AR+BR
CR=CR+1, BR=BR+1
AR=AR-CR

Q2. What do you mean by General register organization? Draw the block diagram of bus organization of GRO having 7 CPU registers.

Q3. Perform the arithmetic operations (+42)+(-13) and (-42)–(-13) in binary using signed-2's complement representation for negative numbers.

Q4. What is hardwired and microprogrammed control unit? Write the control signals for executing the complete instruction XOR R1,R2.

Q5 What do you mean by write-through and write-back? A four-way set associative cache memory uses block of eight words. The cache can accommodate a total of 2048 words from memory. The memory size is 256Kx32. Formulate all the pertinent information for required to construct the cache memory. What is the size of cache memory including valid bit?

Q6. Explain the following:

a. System Bus
b. Memory mapped I/O
c. Isolated I/O
d. Interrupt

Section- B

Note: Attempt ALL Questions. 5 x 3 = 15 marks

Q1. What is instruction cycle? Draw the flowchart for instruction cycle.

Q2. What do you mean by high-impedance state of buses? A computer uses RAM chips of 1024x1 capacity. How many chips are needed to provide a memory capacity of 1024 bytes? How many address lines are required to access 2048 bytes of memory?

Q3. Why I/O Interface is needed? Explain the various I/O commands.

Q4. What is bus arbitration? Differentiate strobe control and handshaking by using suitable figure.

Q5. What is the use of branch instruction? Write the sequence of control steps for executing an unconditional branch instruction.

Section- C

Note: Attempt ANY THREE Questions. 3 x 5 = 15 marks

Q1. What do you mean by pipeline processing? In a certain scientific computation, it is necessary to perform the arithmetic operation Ai*Bi*Ci+Di+Ei with a stream of numbers. Specify a pipeline

configuration to carry out this task. List the content of all the registers in the pipeline for i=1 to 4.

Q2. What do you mean by locality of reference? A computer employs RAM chips of 256x8 and ROM chips of 1024x8. The computer system needs 2K bytes of RAM, 4K bytes of ROM, and four interface units each with four registers. A memory-mapped I/O configuration used. The two highest-order bits of the address bus are assigned 00 for RAM, 01 for ROM, and 10 for interface registers.

a. How many RAM and ROM chips are needed.
b. Give the address range in hexadecimal for RAM, ROM and interface.

Q3. Differentiate programmed I/O and interrupt initiated I/O. Explain in detail the Direct Memory Access (DMA) transfer in a computer system by giving appropriate figure.

Q4. What do you mean by virtual memory? How it is useful for executing any program. A virtual memory system has an address space of 8K words, a memory space of 4K words, and page and block sizes of 1K words. The following page reference changes occur during a given time interval.

4 2 0 1 2 6 1 4 0 1 0 2 3 5 7

Determine the four pages that are resident in main memory after each page reference change if the replacement algorithm used is FIFO and LRU.

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End Term Examination, Odd Semester 2021-22
B.Tech (CSE), IInd Year, IIIrd Semester

BCSC 0009: Software Engineering

Time: 3 Hour **Maximum Marks: 50**

Section- A

Note: Attempt ANY FIVE Questions. 5 x 4 = 20 marks

(I) Explain Agile Software Development model with the help of diagram.

(II) Consider a Pizza Delivery system. Where client can order a pizza on pizza App. Pizza company collect all the orders, prepare the Pizza, deliver the Pizza, receive the payment from client and give payment to his employee's.
Draw DFD level-0, level-1 by considering above scenario.

(III) An authorized User can withdraw or deposit the money using an ATM machine. Create sequence diagram by showing all the necessary steps of ATM machine. Also discuss the purpose of creating sequence diagram.

(IV) What is the difference between Reverse Engineering and Re-engineering? Explain with diagram.

(V) Consider a program for the determination of the nature of roots of a quadratic equation. Its input is a triple of positive integers (say a,b,c) and values may be from interval [0,100]. The program output may have one of the following words.
[Not a quadratic equation; Real roots; Imaginary roots; Equal roots]
Design the Robust test cases.

(VI) What is the use of Structure chart in system design? Create Structure chart by considering the following code:

```
dim num1, num2 as integer

sub calculateAverage()
 dim avg as integer
```

```
inputNums()
avg = average(num1, num2)
outputAvg(avg)
end sub

function average(a,b)
 return (a + b) / 2
end function

sub inputNums()
 num1 = console.readline()
 num2 = console.readline()
end sub

sub outputAvg(x)
 console.writeline("average = " & x)
end sub
```

Section- B

Note: Attempt ALL Questions. 5 x 3 = 15 marks

(I) What is Module Cohesion? Explain any five Cohesion types.

(II) Consider a program to find Largest number among three numbers. Design the test cases using Decision Table Based Testing.

(III) Design a flow chart for Test Driven Development Coding process.

(IV) Explain the following:
(a) Modularization Curve (b) Module Coupling (c) Unit Testing

(V) Let us consider an example of software application. There is function of software application that accepts only particular number of digits, not even greater or less than that particular number.
Consider an OTP number that contains only 6 digit number, greater and even less than six digits will not be accepted, and the application will redirect customer or user to error page. If password entered by user is less or more than six characters, that equivalence partitioning method will show an invalid OTP. If password entered is exactly six characters, then equivalence partitioning method will show valid OTP.
Design the classes and test cases using equivalence class testing technique.

Section- C

Note: Attempt ANY THREE Questions. 3 x 5 = 15 marks

(I) Explain SEI- CMM Model of software quality. Define any five important parameters to judge the quality of a software.

(II) Explain Taute Maintenance Model with the help of diagram.
A software project has development effort of 500 PM. 90 KLOC is added and 60 KLOC is deleted from of software due to maintenance work per year. Some of the cost multipliers are given as:
1. Required software Reliability (RELY) : high= 1.15
2. Date base size (DATA) : high= 0.86
3. Analyst capability (ACAP) : high= 0.82
Other multipliers are nominal. Calculate the Annual Maintenance Effort (AME). size of the project is 600KLOC.

(III) Consider the program(given in below Question IV). List out the operators and operands and also calculate the values of software science measures like η, N, V.

(IV) Draw the Control flow graph for the given program. Find out all independent paths and Design the test cases that will guarantee that all statements in the program have been tested. Also verify your answer using cyclomatic complexity.

```
#include <stdio.h>
void main()
{
  int marks;
  printf("Enter your marks?");
  scanf("%d",&marks);
  if(marks > 85 && marks <= 100)
  {
    printf("Congrats ! you scored grade A ...");
  }
  else if (marks > 60 && marks <= 85)
  {
    printf("You scored grade B + ...");
  }
  else if (marks > 40 && marks <= 60)
  {
    printf("You scored grade B ...");
```

```
    }
    else if (marks > 30 && marks <= 40)
    {
      printf("You scored grade C ...");
    }
    else
    {
      printf("Sorry you are fail ...");
    }
}
```

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End Term Examination, Odd Semester 2021-22
B. Tech, AIML, Year-II, Semester-III
BCSE 0701: Introduction To Machine Learning

Time: 3 Hour **Maximum Marks: 50**

Section- A

Note: Attempt ANY FIVE Questions. 5 x 4 = 20 marks

(I) Given the confusion matrix, find: Classification Accuracy, Recall, Precision, F-measure.

N= 165	Predicted: No	Predicted: Yes	
Actual : No	50	10	60
Actual : Yes	5	100	105
	55	110	

(II) Write the hypothesis, parameters, and cost function of linear regression with multiple variables.

(III) What are ensemble methods? Differentiate between bagging and boosting ensemble methods.

(IV) How to avoid the overfitting in machine learning? Explain it in detail.

(V) What is the Curse of Dimensionality? Demonstrate the projection of high dimension to low dimension. Discuss the steps of the PCA algorithm with an example.

(VI) What are the different layers on CNN? Differentiate between ANN and CNN.

Section- B

Note: Attempt ALL Questions. 5 x 3 = 15 marks

(I) What is hierarchical clustering? Explain its types.

(II) How to decide the optimal number of K in the K-means Algorithm?

(III) Briefly outline the major ideas of naive Bayesian classification in machine learning.

(IV) What methods are used to handle non-linear decision boundaries in SVM? Enlist two constraints on weight (**w**) and constant (**b**) of the line in the middle of the maximum margin between support vectors.

(V) Enlist atleast one method of feature selection and feature extraction and differentiate them.

Section- C

Note: Attempt ANY THREE Questions. 3 x 5 = 15 marks

(I) Suppose you are given the following positively labeled data points $\left\{\binom{3}{1}, \binom{3}{-1}, \binom{6}{1}, \binom{6}{-1}\right\}$ and following negatively

labeled data points, $\left\{\binom{1}{0}, \binom{0}{1}, \binom{0}{-1}, \binom{-1}{0}\right\}$. Plot the graph for the positively and negatively labeled data points, identify the support vectors and compute the equation of hyperplane using Support Vector Machine.

(II) Suppose there are several objects (4 types of medicines) and each object has two attributes or features as shown in the table below. Apply the K-means clustering algorithm to group these objects into the K=2 group of medicine based on the two features (pH and weight index).

Object	X: Weight index	Y:pH
Medicine A	1	1
Medicine B	2	1
Medicine C	4	3
Medicine D	5	4

(III) Compute the weighted sum of Gini Index for 'Past Trend' using the following dataset for Decision Tree:

Past Trend	Open Interest	Trading Volume	Return
Positive	Low	High	Up
Negative	High	Low	Down
Positive	Low	High	Up
Positive	High	High	Up
Negative	Low	High	Down
Positive	Low	Low	Down
Negative	High	High	Down
Negative	Low	High	Down
Positive	Low	Low	Down
Positive	High	High	Up

(IV) Suppose you have given a dataset of weather conditions and corresponding target variable "Play". So, using this dataset you need to decide that whether we should play or not on a particular day according to the Temperature using Naïve Bayes Classifier.

Problem statement: "If the temperature is Hot, then the Player should play or not?"

Temperature	Play
Hot	No
Hot	No
Hot	Yes
Mild	Yes
Cool	Yes
Cool	No
Cool	Yes
Mild	No
Cool	Yes
Mild	Yes
Mild	Yes
Mild	Yes
Hot	Yes
Mild	No

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End Term Examination, Odd Semester 2021-22

Program-B.Tech, Branch-CCV, Year-II, Semester-III

Subject Name: Introduction to Virtualization and Cloud Computing

Subject Code: BCSE1502

Time: 3 Hour **Maximum Marks: 50**

Section- A

Note: Attempt ANY FIVE Questions. 5 x 4 = 20 marks

(I) What arrangements an organization need to do while migration from conventional IT server or datacenter to virtualized environment?

(II) Suppose you are a Storage Administrator of an organization that has offices in different region of a country. So, how you manage storage capacity of the organization?

(III) Identify the types of virtualization an organization must use to set up its offices in different part of the world. Justify your answer with proper reasoning. Also, discuss the cost impacts of proposed virtualization.

(IV) Explain with the help of example, how cloud computing is related to client server model, cluster computing, grid computing and utility computing?

(V) What technique is used to reduce the broadcast domain of a network? Explain with the help of example.

(VI) What are the different features of cloud computing? Discuss the various concerns included in CAPEX and OPEX.

Section- B

Note: Attempt ALL Questions. 5 x 3 = 15 marks

(I) Compare single tenant architecture with multi-tenant architecture. How resource pooling increases performance,

utilization & serviceability?

(II) Discuss the different components of PaaS. Also, what things you need to consider before choosing a Paas provider?

(III) Discuss the benefits and risks associated with each cloud deployment model.

(IV) Virtual servers run on a hypervisor, and hardware resources are emulated for the virtual servers via the hypervisors. If the hypervisor fails or if the underlying physical server fails (thereby causing the hypervisor to fail), the failure condition cascades to all of its hosted virtual servers. Design a cloud architecture to overcome this failure.

(V) When a cloud consumer chooses what IT resources it would like to lease, having the actual provisioning of these IT resources performed manually can require too much time and human interaction to be sufficiently effective and responsive. Develop a model to show how can the provisioning of IT resources be automated and made available to cloud consumers on-demand?

Section- C

Note: Attempt ANY THREE Questions. 3 x 5 = 15 marks

(I) Analyze the decision factor matrix for different cloud deployment models based on the following parameters: Business IT control, business critical application, data and transaction security, balance of capex and opex, workload characteristics, workload lifespan, industry segment, data freedom, and time to deploy.

(II) It is challenging to equip an IT resource to match its processing requirements. If the demand for the IT resource is below its capacity, then it is under-utilized and if the demand is above its

capacity, it is over-utilized or unable to meet the demand. Develop a service model to show how can IT resources be scaled automatically in response to fluctuating demand?

(III) Classify the Microsoft office 365 in one of the cloud computing delivery models. Also, analyze the advantages and disadvantages of the using the same delivery model in respect to end users.

(IV) Assume that you are a GLA University administrator and you have to decide which deployment model is required for our organization to establish cloud-based web services for end user (students and faculty). You have to critically analyze which deployment model provides stringent security to setup services and why? You have to also detailed out the steps for virtualization of existing data center in GLA University.

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End Term Examination, Odd Semester 2021-22

B.Tech. CSF II Year, III Semester

BCSE0601: IT Application and Data Security

Time: 3 Hour **Maximum Marks: 50**

Section- A

Note: Attempt ANY FIVE Questions. 5 x 4 = 20 marks

Q1. Discuss Data tampering vulnerability. Explain how to protect Data tampering in URL and Protection in database.

Q2. What do you understand by input validation? Discuss some of the best practices that can help mitigate input validation attack.

Q3. Define Cookie replay attack. Discuss the systematic protection and mitigation method for cookie replay attack.

Q4. Explain in detail Auditing and assurance mechanisms with the help of a suitable example.

Q5. Describe how a man-in-the-middle attack may be performed on a Wi-Fi network and what is the consequences of such an attack?

Q6. What do you understand by Auditing & Logging? Explain any Four strategic countermeasures for protecting web applications.

Section- B

Note: Attempt ALL Questions. 5 x 3 = 15 marks

Q1. Why Data security is extremely important for all companies? Explain in which technique values and actions are inserted unexpectedly as an application's input to crash it down.

Q2. What do you understand by network based threats? Discuss the various data threats with example.

Q3. What do you understand by Pharming? Discuss and explain Domain Name System Security Extension(DNSSEC) is an extension of DNS that provides three distinct services.

Q4. What is Cryptanalysis. Discus and explain discuss various Cryptanalysis attack.

Q5. Discuss database threat technique. Explain various Database Security Countermeasures.

Section- C

Note: Attempt ANY THREE Questions. 3 x 5 = 15 marks

Q1. Why RSA algorithm is so useful? Encrypt and decrypt the message "cryptography" by using by using RSA algorithm with prime numbers p=5 and q=7. Use encoding 00 to 25 for letters A to Z respectively.

Q2. What do you mean by TCP/IP? Which layers of TCP/IP can be used to data security. Discuss in detail the different mechanism used to provide security at different layers of TCP/IP.

Q3. Write the steps of Elgamal digital signature scheme. Using the elgamal signature, let prime number is q=19, primitive root α=10, private key XA=15, Random integer K=5.find the value of signature S1 and S2 if h(M)=14.verify the signature.

Q4. Write short note on the following:

1) Wireless network security threats
2) Physical security threats
3) Network based threats
4) Web application threats
5) Data security threats

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End Term Examination, Odd Semester 2021-22

B.Tech(CSE(DA)), IIndYear,IIIth Semester

BCSE0551:Introduction to Business Analytics

Time: 3 Hour **Maximum Marks: 50**

Section- A

Note: Attempt ANY FIVE Questions. 5 x 4 = 20 marks

(I) Black boxes used in aircrafts are manufactured by three companies A, B and C. 75% are manufactured by A, 15% by B, and 10% by C. The defect rates of black boxes manufactured by A, B, and C are 4%, 6%, and 8%, respectively. If a black box tested randomly is found to be defective, what is the probability that it is manufactured by company A?

(II) Define business analytics. List down the importance of business analytics.

(III) Discriminate among predictive, perspective and descriptive analytics in tabular form.

(IV) Define stratified random sampling and write the advantages of stratification.

(V) List down the steps that are followed during any sampling process.

(VI) If a patient is waiting for a suitable blood donor and the probability that the selected donor will be a match is 0.2, then find the expected number of donors who will be tested till a match is found including the matched donor.

Section- B

Note: Attempt ALL Questions. 5 x 3 = 15 marks

(I) Let's say that 80% of all business startups in the IT industry report that they generate a profit in their first year. If a sample of 10 new IT business startups is selected, find the probability that exactly seven will generate a profit in their first year.

(II) The average number of homes sold by the Acme Realty company is 2 homes per day. What is the probability that exactly 3 homes will be sold tomorrow?

(III) The lifetime T (years) of an electronic component is a continuous random variable with a probability density function given by

$$f(t) = e^{-t} \quad t \geq 0 \quad \text{(i.e. } \lambda = 1 \text{ or } \mu = 1\text{)}$$

Find the lifetime L which a typical component is 60% certain to exceed. If five components are sold to a manufacturer, find the probability that at least one of them will have a lifetime less than L years.

(IV) List down various types of non-probability sampling technique. Also describe briefly.

(V) Define Data Warehouse. List down different characteristics of data warehouse.

Section- C

Note: Attempt ANY THREE Questions. 3 x 5 = 15 marks

I. What is Big Data Analytics? Discuss two use case of Big Data Analytics with their Brief scenario. Discuss various characteristics of Big data that helpful in discriminating with other data.

II. Discuss CRISP-DM? What is Data mining? What are the types of Machine Learning Algorithms are there? Discuss them with example.

III. A sample of 100 patients was chosen to estimate the length of stay (LoS) at a hospital. The sample mean was 4.5 days and the population standard deviation was known tobe 1.2 days.

   a. Calculate the 95% confidence interval for the population mean.
   b. What is the probability that the population mean is greater than 4.73 days?

IV. Compute Pearsons coefficient of correlation between advertisement cost and sales as per the data given below:

Advertisement Cost in 1000's	39	65	62	90	82	75	25	98	36	78
Sales in lakhs	47	53	58	86	62	68	60	91	51	84

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End-Term Examination, Odd Semester 2021-22

B.Tech. CSE (IIOT), II: Year, III: Semester

BCSE 0902: Smart Industrial Connectivity

Time: 1.5-hour Maximum Marks: 50

Section A

Note: Attempt all questions. 30 x 1 = 30 Marks

1. LAN speed supported by Raspberry pi 3:
   A. 10mbps
   B. 100mbps
   C. 1000mpbs
   D. All of the above

2. Which of the device helps in getting coordinates for latitude and longitude?
   A. GSM
   B. GPS
   C. RFID
   D. None of the above

3. Python files are saved in which format?
   A. .txt
   B. .py
   C. .pi
   D. .exe

4. CD command is used to:

A. Change directory
B. Current directory
C. Change data
D. Current data

5. GPIO.BCM library is used to access:
   A. Wi-Fi
   B. Bluetooth
   C. Broadcom SOC channel
   D. RAM

6. Raspbian OS kernel is based on which platform?
   A. Windows
   B. Linux
   C. Solaris
   D. Mac

7. MISO in RFID stands for:
   A. Master out slave in
   B. Master in slave out
   C. Master in slave in
   D. Master out slave out

8. What is the read range of low frequency MFRC522 RFID tag:
   A. 1cm
   B. 2cm
   C. 5cm
   D. 10cm

9. Video output type supported by raspberry pi 3?

A. HDMI
B. Optical
C. USB
D. Micro HDMI

10. Type of audio jack supported in raspberry pi3?
A. 1.0mm
B. 2.5mm
C. 3.5mm
D. none of the above

11. The .....command moves a file and places it at the specified location
A. move
B. mv
C. mov
D. mve

12. You can use ..... to list the contents of file(s)
A. cat
B. ct
C. cts
D. cnt

13. The ----- command displays the beginning of a file.
A. head
B. hds
C. hde
D. heads

14. ------ denotes the secure shell. Connect to another computer using an encrypted network connection.

A. sshe
B. sshs
C. ssh
D. sesh

15. The ------ command enables you to run a command as a superuser, or another user.

A. suddo
B. su-do
C. sudo
D. sudos

16. --------- command is used to check that whether communication can be made with another host or not.

A. pin-g
B. Ping
C. Pi-ng
D. pings

17. ----------command will tell you the network configuration details such as your IP address.

A. ifconfig
B. ipconfig
C. ip-config
D. if-config

18. ---- command is used to install, upgrade and remove the packages.

A. apt-get
B. aptget
C. aptgets

D. aptgets

19. ------- command will open the configuration settings menu.
A. raspiconfig
B. raspi-config
C. ras-piconfig
D. raspicon-fig

20. ----- Upgrades all of the software packages you have installed.
A. apt-gets upgrade
B. apts-get upgrade
C. aptget upgrade
D. apt-get upgrade

21. What does sudo reboot does
A. Shut down
B. File open
C. Restart
D. None of them

22. Which pins are used for SPI connection with RPi interfacing with RFID?
A. RX and TX
B. MISO and MOSI
C. SS, SCK, MISO and MOSI
D. SDA and SCL

23. Is Python case sensitive when dealing with identifiers?
A. Yes
B. No

24. Which of the following data types is not supported in python?

A. String
B. Numbers
C. Slice
D. List

25. Select the proper time delay given in the following line of code:
time.sleep(0.1)

A. 100 mS
B. 10 mS
C. 1 mS
D. 0.1 mS

26. Raspberry Pi3 is consists of an inbuilt Bluetooth system.

A. True
B. False

27. How many GPIO pins RPi 3B have?

A. 24
B. 18
C. 28
D. 22

28. Name the communication interface used in Raspberry Pi 3?

A. UART
B. SPI
C. I2C
D. All of them

29. What is the Processor operating voltage of RPi3?

A. 3.3V
B. 4.1V
C. 5.2V
D. 3.2V

30. What is the raw voltage input given to RPi3?

A. 5V,2A
B. 5V,3A
C. 2V,5A
D. 4V,4A

Section B

Note: Attempt all questions. 40 x 0.5 = 20 Marks

31. Find the perfect match:
GPIO.setmode(GPIO.BCM)
Pin = 4
Here Pin = 4 will represent:

A. GPIO PIN
B. Physical Pin

32. How many analog pins are available in RPi.

A. 1
B. 2
C. 3
D. None of the above

33. Select the property used to fetch the data to Mashup widgets.

A. GetProperties
B. SetProperties

C. GetsProperties
D. Get-Properties

34. Thingworx is the product of ------- company.
A. PTTC
B. PTC
C. DASSAULT
D. RENOULT

35. 401 HTTP response stands for:
A. Unauthorized access
B. Page not found
C. Method not allowed
D. None of above

36. Select the right function to read humidity and Temperature using DHT11 in Rpi.
A. read_dht11
B. read_retry
C. read_try
D. None of the above

37. If input to the Voltage Sensor is 22 V using voltage divider of R1 = 30 Kohm & R2 = 7.5Kohm, what will be the appeared potential at the output terminal.
A. 4.9v
B. 4.4v
C. 4.3v
D. 4.88v

38. Voltage Sensitivity of LM35 is:
A. 100mV/ °C
B. 10mV/°C
C. 12mv/°C
D. 160 mV/°C

39. Type of communication protocol(s) supported by MFRC522 RFID?
A. UART
B. UART and I2C
C. UART, I2, and SPI
D. None of the above

40. Select the function used to return the Timestamp of Rpi.
A. time.time()
B. time_time()
C. time as time()
D. time.tm()

41. ----------Synchronizes the list of packages on your system to the list in the repositories.
A. apt-gets update
B. apts-get update
C. aptget update
D. apt-get update

42. ---------- To shut down immediately.
A. poweroff
B. power-off
C. pwroff
D. shdwn

43. --------- Prints the current date.

A. date

B. dt

C. date-

D. date-h

44. What is the processing speed of the Raspberry pi 3 model.

A. 700 mhz

B. 850 mhz

C. 1.2 ghz

D. 1.4 ghz

45. python prompt is represented as

A. >

B. >>

C. >>>

D. $

46. _____ is the example of Passive RFID tag.

A. Metro token

B. Fast-tag

C. MFRC522-Tag

D. All of the above

47. REST stands in RESTful HTTP method:

A. Representational State Transfer

B. Representational Status Transfer

C. Representational State Transmission

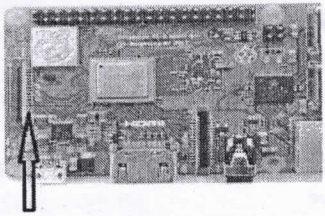
D. None of the above

48. ------Successful server response of HTTP protocol:
A. 301
B. 400
C. 399
D. 200

49. Which out of below option is not a property parameters of Thing:
A. Logged
B. Persistent
C. Read Only
D. TimeStamp

50. Name the service used to access historical data from database.
A. QueryPropertyHistory
B. QueryRequestHistory
C. QuerySetProperty
D. None of the above

51. The Arrow pointing Part of the RPi stands for?



A. Ethernet controller
B. DSI Display port
C. Video Port
D. Dual band WIFI

52. DNS stands for .....
A. determine name system
B. domain name system
C. device name system
D. development name system

53. RISC stands for .....
A. reduced instruction set architecture
B. rare information set architecture
C. reduce information set architecture
D. rare instruction set architecture

54. API stands for .....
A. application programming interface
B. Android programming interface
C. Arduino protocol information
D. application protocol interface

55. Why requests library is used? Select that applies:
A. Standard for making HTTP requests in Python
B. Provides a clear and concise framework to work with HTTP requests.
C. Abstracts the complicated parts of HTTP protocol.
D. None of the above

56. 404 response code for a HTTP request stands for:
A. Page not found
B. Unauthorized Access
C. Method not allowed
D. None of the above

57. "requests.put() function is used to:

A. Receive a data
B. Send a data
C. Both of the above
D. None of the above.

58. Select the temperature range measured by LM35:

A. -20 °C to 100 °C
B. -55 °C to 150 °C
C. -10 °C to 150 °C
D. -20 °C to 150 °C

59. Select the voltage range that can be measured by Voltage sensor:

A. 0 V – 10 V
B. 0 V – 25 V
C. 0 V – 12 V
D. 0 V – 9 V

60. Select the correct formula for voltage divider Circuit:

A. Vout = Vin (R2 / (R1 + R2))
B. Vout = Vin ((R1 + R2) / R2)
C. Vout = Vin (R1 / R2)
D. None of the above

61. To start Python from the command prompt, use the command_____.

A. Execute python
B. Go python
C. python
D. run python

62. Which of the following functions is a built-in function in python language?

A. val()
B. print()
C. printls()
D. None of these

63. Study the following function:

>>> round(4.576)

What will be the output of this function?

A. 4
B. 5
C. 576
D. 5

64. Study the following statement:

>>>"a"+"bc"

What will be the output of this statement?

A. a+bc
B. abc
C. a bc
D. a

65. Choose the command for file system from following in Rpi system:

A. ls-
B. lst-s
C. ls
D. l-s

66. Which one of the following is not an HTTP Method:

A. DELETE

B. GET
C. POST
D. UNDO

67. ------- command displays the name of the present working directory:
A. pwds
B. pwd
C. pdw
D. pw-d

68. You can use ---------to create a new directory:
A. makdir
B. mkdir
C. mkdirs
D. mkdirec

69. To remove directories, use .....
A. rmdr
B. rem-dir
C. remdir
D. rmdir

70. Using ..... makes a copy of a file and places it at the specified location:
A. cpi
B. cop
C. cpy
D. cp

Set B

Printed Pages: 12 **University Roll No.....**

End-Term Examination, Odd Semester 2021-22

B.Tech (All Branches), Second Year, III Semester

English for Professional Purposes I: BELH0003

Time :3 Hours Maximum Marks: 50

Communication Skills **[TIME: 90 MINUTES]**

Instructions:

1. It is mandatory to clearly mention your set number in the answer sheet.
2. Word limit for the questions in Section B is 200.

Section- A **2x4 = 8Marks**

Attempt *any two* questions.

1. Identify and mark word stress in the following words:

A. communication	B. democracy	C. democratic	D. geology
E. dependability	F. present (noun)	G. present (verb)	H. photography

2. In the following passage, identify (1) the most likely thought group boundaries, and (2) the focus word in each thought group.

Everyone values the importance of agriculture in the economy of India. Universities and research institutions have been established to find ways to improve output. Union and state governments have ministries and departments of agriculture. Its place in economy is unquestionable.

3. Identify the number of syllables in the following words:

A. identify	B. cancellation	C. computer	D. elucidate
E. meter	F. marriage	G. Professional	H. agenda

Section- B **2x6 = 12 Marks**

Attempt ***any two*** questions.

4. Discuss organizational barriers to communication in detail and suggest ways to overcome them.
5. Write an essay on any one of the following:
   A. Is the colonial mentality hindering India's success?
   B. Digital economy: A leveller or a source of economic inequality?
6. What kind of phrases/sentences/expressions should you use in the following situations in a group discussion?
   A. Initiating
   B. Agreeing
   C. Disagreeing

Attempt all questions.

1.___ July 2022, we have to finish the project.

A. by
B. at
C. on
D. from
E. since

2. Ruksana teaches piano every day ___ Saturday and Sunday.

A. with
B. on
C. except
D. in
E. at

3. Iqbal wants an apartment which has windows facing ___ the sea.

A. to
B. no preposition required
C. on
D. in
E. at

4. After the head injury the doctor advised Latif ___ travelling.

A. except
B. of
C. to
D. against
E. from

5. It was very kind ___ my colleagues that they prepared food for me in my absence.

A. for
B. from
C. in
D. with
E. of

6. The deadline is 1st January. We have to finish the assignment ___ or before that day.

A. in
B. at
C. from
D. up
E. on

7. From seven o'clock ___ nine, the restaurant will be open.

A. at
B. in
C. on
D. from
E. till

Q-Read each sentence given below and find out whether there is an error in it. The error, if any will be one of the parts of the sentence which are marked as a, b, c, d and e. If there is no error, the answer will be (e) i.e. No error.

8.Divya in addition to her friends(a)/ accused of not (b) /telling the truth(c) / in the court of law(d) /no error (e).

A. a
B. b
C. c
D. d
E. e

9. The number of accidents (a)/ are increasing(b)/ day by day(c)/ on the Delhi metro route(d)/ no error (e).

A. a
B. b
C. c
D. d
E. e

10. Either my colleague (a)/ or a peon (b)/ are coming home(c) with the material today (d)/ No error (e).

A. a
B. b
C. c
D. d
E. e

Q. Choose the correct options to fill in the blanks.

11. They ..... to welcome new CEO of the organization.

A. standing up
B. astound up
C. stood up
D. to stand
E. to stood

12. While leaving for office, I ..... keys on the table.

A. saw
B. seen
C. see
D. had see
E. has seen

13..... a coconut tree to pluck coconuts is a thrilling experience."

A. "To climb
B. "To climbed
C. " To Climbing
D. "Climbed
E. "Climbing

14. I don't see why you .....borrow his cycle. You have one of your own.

A. should
B. could
C. can
D. may
E. might

15. Farmers ..... this land and this must be used only for cultivation.

A. owns
B. owning
C. own
D. has owned
E. have been owning

16. What kind of a verb is the underlined word in the given sentence?

Ice skating is my favourite sport.
A. Gerund
B. Participle
C. Infinitive
D. Bare infinitive
E. Finite verb

17. What kind of a word is the underlined one in the given sentence?
"The broken mirror reflected the images in a distorted manner."
A. Gerund
B. Participle
C. Infinitive
D. Bare infinitive
E. Finite verb

18.What kind of a word is the underlined one in the given sentence?

"Offering the fruits of action to the Lord, a karmayogi attains everlasting peace in the form of God-realization."
A. Gerund
B. Participle

C. Infinitive
D. Bare infinitive
E. Finite verb

19.What kind of word is the underlined one in the given sentence?

He enjoys swimming in the sea.
A. Gerund
B. Participle
C. Infinitive
D. Bare infinitive
E. Finite verb

In the following passage choose from the choice of five words one which fits the meaning of the passage.

All soil is not the same. It has varying textures. (20)..... describe soil based on the (21)..... of clay, silt, and sand (22)..... contains. The combination of these (23) ..... things affects the way the (24) ..... looks and feels. It also (25) ..... its ability to help plants (26) ..... Clay particles are the smallest (27) ..... in soil. They can hold (28) ..... lot of nutrients. Clay particles (29) ..... smooth when they are dry (30)..... very sticky when wet. Water (31)..... easily move through clay particles.(32) ..... can't, either. Wet clay is (33) .... sticky; even water and air (34) ..... stuck! Silt particles are a (35).... bigger than clay particles. (36). ..... have a powdery texture and (37)..... smooth.
Sand particles are the (38) ..... particles in the soil. Sand helps (39) ..... break up other soil particles so that air and water can move through the soil more easily.

Pick the correct answer from the given options-

20. A. Scientist
B. Scientists
C. Teacher
D. Engineer
E. Nehru

21 A. amount
B. quality
C. weight
D. measure
E. kilograms

22.A. she
B. he
C. it
D. you
E. none of the above

23.A. three
B. there
C. here
D. four
E. None of the above

24.A. soil
B. water
C. clay
D. muddy
E. all of the above

25.A. affects
B. effects
C. apex
D. affect
E. effect

26.A. Growing
B. Grow
C. Grown
D. Grew
E. None of the above

27.A. particles
B. particle
C. article
D. articles
E. bit

28.A. this
B. these
C. that
D. a

E. those

29.A. are
B. is
C. am
D. being
E. are being

30.A. but
B. moreover
C. nevertheless
D. beneath
E. if

31.A. can
B. oughtn't
C. must not
D. cannot
E. will

32.A. river
B. water
C. sunlight
D. rain
E. air

33.A. so
B. when
C. very
D. a lot
E. many

34.A. get
B. be
C. becomes
D. being
E. may

35.A. much
B. very
C. many
D. time
E. little

36.A. There
B. Their
C. Them
D. They
E. Theirs

37.A. feels
B. felt
C. feel
D. feeling
E. fell

38.A. large
B. largest
C. larger
D. long
E. elongated

39.A. too
B. will
C. to
D. many
E. all of the above

The first sentence is fixed. Choose the most logical order of sentences from among the given choices to construct a consistent, logical paragraph.

40. It was an expected coronation but the manner in which it transpired seemed like an exercise in smoke and mirrors.

A. But the breaking-news bit was reserved for the last line, which stated that Rohit Sharma would be the captain of the ODI and T20I teams.

B. It was a press release from the Board of Control for Cricket in India (BCCI) and almost its entire part was devoted to the naming of the Test squad under Virat Kohli, set for the tour of South Africa.

C. After Kohli relinquished the T20I captaincy, Rohit had already taken over the reins in the shortest format and it was a matter of time before he was given charge of the One Day International squad too.

D. Late on Wednesday evening, an email popped up in the inboxes of Indian cricket writers.

A. BCDA B. CBDA C. DBAC D. BDAC E. ACDB

Choose the most logical order of sentences from among the given choices to construct a consistent, logical paragraph.

41.A."This," he added, "will be one of the proudest moments in Indian history."

B. The room was fully packed with journalists and photographers.

C. It was a tiresome evening and I was prepared to leave when my editor ran in.

D. "Gentlemen, it is an honour to tell you that in 2025, ISRO is going to launch its first manned-spacecraft for Mars," he started.

E. We were busy talking when suddenly the Prime Minister entered the press conference room.

F. He told me that a fax had come in informing about a press conference to be held at the Prime Ministerial office and that I need to be there as soon as possible.

A. CDBFEA B. CDBAEF

C. DBCAFE D. DBCAEF

E. CFBEDA

Choose the most logical order of sentences from among the given choices to construct a consistent, logical paragraph.

42. A."This," he added, "will be one of the proudest moments in Indian history."

B. The room was fully packed with journalists and photographers.

C. It was a tiresome evening and I was prepared to leave when my editor ran in.

D. "Gentlemen, it is an honour to tell you that in 2025, ISRO is going to launch its first manned-spacecraft for Mars," he started.

E. We were busy talking when suddenly the Prime Minister entered the press conference room.

F. He told me that a fax had come in informing about a press conference to be held at the Prime Ministerial office and that I need to be there as soon as possible.

A. CDBFEA B. CDBAEF

C. DBCAFE D. DBCAEF

E. CFBEDA

Choose the most logical order of sentences from among the given choices to construct a consistent, logical paragraph. The first sentence is fixed.

43.Drugs are big business in Afghanistan, and under the Taliban, trade is booming.

A. The rise of meth has been fuelled by a discovery that ephedra - a common, wild herb known locally as "oman" - can be used to make one of the drug's key ingredients: ephedrine.

B. Previously, the Taliban were understood to charge tax on ephedra.

C. The country has long been linked with heroin, but in recent years, it has also emerged as a significant producer of crystal meth - another dangerously addictive drug.

D. But recently, they have announced a ban on its cultivation, in a decree that was not widely publicised.

E. One source involved in the trade says that about 3,000kg of crystal meth are now manufactured every day by more than 500 makeshift "factories" in a single remote drug-producing district in the south-west of the country.

F. At a bazaar deep in the desert that serves as the central node of Afghanistan's meth trade, there are huge mounds of the plant on sale, on a scale not previously seen before.

A. CBDFEA B. CEAFBD
C. CEADBF D. CEDAFB
E. CEADFB

Choose the most logical order of sentences from among the given choices to construct a consistent, logical paragraph.

44. A. His father was a lieutenant general in the Indian Army and his mother was the daughter of a politician.
B. General Bipin Rawat, who died in a helicopter crash on Wednesday, was India's most senior defence official.
C. He was appointed as the country's first chief of defence staff (CDS) in 2019, and previously served as the chief of India's one-million strong army.
D. Gen Rawat was born on 16 March 1958 in the northern Indian state of Uttarakhand.
E. The 63-year-old had the reputation of being a tough soldier and an inspiring commander, who sometimes sparked controversy with his comments on political developments.

A. BCEDA B. BCDEA
C. CEDAB D. ADBEC
E. ADBCE

Choose the most logical order of sentences from among the given choices to construct a consistent, logical paragraph.

45. A.But the true number is likely to be far higher.
B.With two doses of a vaccine "not enough" Mr Zahawi encouraged people to get a booster jab - those aged 30 and over are eligible to do so from Monday.
C.The first people in the UK are in hospital with Omicron infections, Nadhim Zahawi has said.
D.There were 48,854 new positive Covid cases reported across the UK on Sunday - there have been 360,480 cases in the last seven days.
E.The new variant of coronavirus now accounts for a third of cases in London, the education secretary said.
F.As of Sunday, there have been 3,137 confirmed cases of the Omicron variant in the UK.

A. BDACEF B. CEBFAD
C. BDAFEC D. CEBFDA
E. BDACFE

Choose the most logical order of sentences from among the given choices to construct a consistent, logical paragraph.

46.A. Events intervened, and in the late 1930s and 1940s, Germany suffered from "over-branding".
B. The British used to be fascinated by the home of Romanticism.
C. But reunification and the federal government's move to Berlin have prompted Germany to think again about its image.
D. The first foreign package holiday was a tour of Germany organized by Thomas Cook in 1855.
E. Since then, Germany has been understandably nervous about promoting itself abroad.

A. ACEBD B. DECAB C. BDAEC
D. DBAEC E. EBCDA

Choose the most logical order of sentences from among the given choices to construct a consistent, logical paragraph.

47. A.This is now orthodoxy to which I subscribe — up to a point.

B. It emerged from the mathematics of chance and statistics.
C. Therefore the risk is measurable and manageable.
D. The fundamental concept: Prices are not predictable, but the mathematical laws of chance can describe their fluctuations.
E. This is how what business schools now call modern finance was born.
A. ADCBE B. EBDCA C. ABDCE
D. DCBEA E. DCBAE

Choose the most logical order of sentences from among the given choices to construct a consistent, logical paragraph.

48. A. Passivity is not, of course, universal.
B. In areas where there are no lords or laws, or in frontier zones where all men go armed, the attitude of the peasantry may well be different.
C. So indeed it may be on the fringe of the unsubmissive.
D. However, for most of the soil-bound peasants the problem is not whether to be normally passive or active, but when to pass from one state to another.
E. This depends on an assessment of the political situation.
A. BEDAC B. CDABE C. EDBAC
D. ABCDE E. DBCAE

Choose the most logical order of sentences from among the given choices to construct a consistent, logical paragraph.

49.A. Luckily the tide of battle moved elsewhere after the American victory at Midway and an Australian victory over Japan at Milne Bay.
B. It could have been no more than a delaying tactic.
C. The Australian military, knowing the position was hopeless, planned to fall back to the south-east in the hope of defending the main cities.
D. They had captured most of the Soloman Islands and much of New Guinea, and seemed poised for an invasion.
E. Not many people outside Australia realize how close the Japanese got.
A. EDCBA B. ECDAB C. ADCBE
D. CDBAE E. DCBAE

Choose the most logical order of sentences from among the given choices to construct a consistent, logical paragraph.

50.A. In the west, Allied Forces had fought their way through southern Italy as far as Rome.
B. In June 1944 Germany's military position in World War Two appeared hopeless.
C. In Britain, the task of amassing the men and materials for the liberation of northern Europe had been completed.
D. The Red Army was poised to drive the Nazis back through Poland.
E. The situation on the eastern front was catastrophic
A. EDACB B. BEDAC C. BDECA
D. CEDAB E. DBCAE

Choose the most logical order of sentences from among the given choices to construct a consistent, logical paragraph.

51.A. The likelihood of an accident is determined by how carefully the motorist drives and how carefully the pedestrian crosses the street.
B. An accident involving a motorist and a pedestrian is such a case.
C. Each must decide how much care to exercise without knowing how careful the other is.
D. The simplest strategic problem arises when two individuals interact with each other, and each must decide what to do without knowing what the other is doing.
A. ABCD B. ADCB C. DBCA
D. DBAC E. CADB

Choose the most appropriate answer.

52. We will not write ..... assignment for this project.

A. some
B. a little
C. a few
D. all
E. any

53. I want my ..... friends to join me on my birthday.

A. much
B. any
C. many
D. neither
E. either

54.....student has completed this assignment before time.

A. Neither
B. All
C. Many
D. Some
E. A little

55. In which sentence the word 'blind' is used as a noun?

A. My brother is blind but he can read and speak eight different languages.
B. We should not let our prejudices blind us to the facts of the situation.
C. These blinds are beautiful for this window.
D. His good looks always made her blind to his faults.
E. She is completely blind to her faults.

56. **Many** a flower ..... born to blush unseen.

A. is
B. are
C. were
D. have been
E. are being

57.There is ..... employment in that area.

A. much
B. many
C. too many
D. any
E. few

58.Could you bring me ..... book which I saw in the book fair.

A. a
B. that
C. those
D. these
E. many

59. Would you lend me your car for ..... days?

A. a few
B. little
C. any
D. much
E. one

60. Rahim worked in a hotel ___ the corner.

A. round
B. in
C. on
D. off
E. from

61. While travelling by bus, you can see many *dhabas* ___ the national highways.

A. in
B. into
C. along
D. around
E. over

62. Attending phone calls ___ driving is very dangerous.

A. from
B. in
C. upon
D. while
E. beside

63. We saw Farooq waving his hand __ the football field.

A. off
B. underneath
C. across
D. into
E. onto

64. The last episode of *Stranger Things* will be over ___ about twenty minutes.

A. in
B. on
C. at
D. for
E. during

65. I love everything about Biryani ___ the cardamoms.

A. except
B. since
C. in
D. on
E. to

66. The spaceship was hovering ___ the old Banyan tree of our village.

A. over
B. above
C. on
D. in
E. under

67.___ time immemorial the Earth has been revolving the Sun.

A. in
B. since
C. on
D. at
E. during

Choose the correct answer

68. What would you call a person who has the talent to make a bad situation better or less harmful?

A. An Antiquarian
B. An Ameliorate
C. A Fulcrum
D. A Freelancer
E. A Collegiate

Choose the correct answer

69. What would you call the act of training someone to accept a particular set of beliefs, especially political, religious ones and not consider any other?

A. Impromptu
B. Hypocrite
C. Indoctrinate
D. Egotist
E. Inimitable

Choose the correct answer

70. What would you call the act of writing an experience of meaning someone important or famous that you like?

A. An Agenda
B. A Memo
C. A memoir

D. An Email
E. A notice

Choose the correct answer

71. A person who is not experienced in a job or situation is called
A. novice
B. newbie
C. entrepreneur
D. debut
E. servile

Choose the correct answer

72. One who possesses many talents is called
A. versatile
B. nubile
C. ornate
D. gazette
E. verbose

Fill in the blanks with the correct answers.

73.The Indian Navy has decided to _____ two battleships.
A. simulate
B. reinstitute
C. decommission
D. rehabilitate
E. verbose

Fill in the blanks with the correct answers.

74. His defeat in the world championship led to a long period of gloomy _____.
A. introspection
B. nubile
C. combustion
D. simulation
E. decommission

Fill in the blanks with the correct answers.

75. All this loud noise _____my headache.
A. accept
B. adopt
C. aggravate
D. advice
E. adapt

Fill in the blanks with the correct answers.

76. Do we have the latest _____of the dictionary?
A. addition
B. edition
C. allusion
D. ensure
E. average

Fill in the blanks with the correct answers.

77. The manager and the director have quite an _____ relationship.
A. amicable
B. amiable
C. median
D. sight
E. ascent

Fill in the blanks with the correct answers.

78. Every day he pays his wife a heartfelt_____.
A. complement
B. compliment
C. coarse
D. course
E. compose

Fill in the blanks with the correct answers.

79. She's making a few _____ inquiries with her mother's friends.
A. decide
B. discreet
C. deduct

D. deduce
E. denote

Fill in the blanks with the correct answers.

80. They've taken on some financial _____who's going to sort out all their problems.
A. solution
B. burden
C. relief
D. purveyor
E. inimitable

Choose the correct meaning-

81. Corroborate
A. to confirm
B. to deny
C. to forget
D. to connect
E. to confess

Choose the correct meaning-

82. Elicit
A. something illegal
B. to draw forth or
C. bring out
D. too quick to believe
E. to trust

Choose the correct meaning-

83. Counsel
A. to get together
B. to give comfort
C. to advice
D. to sign off
E. to leave

Choose the correct meaning-

84. Blatant
A. to inhale
B. to exhale
C. to be obvious
D. to be deceptive
E. to be ignorant

Choose the correct meaning-

85. Cereal
A. something arranged in sequence
B. to aid
C. to cooperate
D. a grain from plants
E. to collaborate

Choose the correct meaning-

86. Conscience
A. to comfort
B. to disturb
C. the ability to sense right from wrong
D. to fake something
E. to console

Choose the correct meaning-

87. Credulous
A. too quick to believe someone
B. too quick to speak
C. too quick to think
D. too quick to sleep
E. too quick to cry

88. Nivedita bought these books from book fair and put ..... in _____ bag.
A. them, those
B. its, that
C. them, that
D. these, those
E. this, that

89. Please do not interfere_____ my personal affairs.
A. about
B. in
C. of
D. on
E. with

90. The TV is too quiet. Can you turn it_____ a bit?
A. after
B. off
C. over
D. up
E. down

Printed Pages: 19 University Roll No.....

End-Term Examination, Odd Semester 2021-22

B.Tech. CSE, II: Year, III: Semester

BCSE 0900: Introduction to IOT

Time: 1.5-hour Maximum Marks: 50

Section A

Note: Attempt all questions. 30 x 1 = 30 Marks

1. What does GPIO stand for?
   A. General Purpose Inner Outer
   B. General Purpose Input Output
   C. General Purpose Interested Old
   D. General Process Input Output

2. What does IDE stand for?
   A. In Deep Environment
   B. Integrated Development Environment
   C. Internal Deep Escape
   D. Integrated Desired Environment

3. It starts with a /* and continues until a */ What does this do?
   A. Loads a sketch
   B. Makes comments
   C. Compiles quicker
   D. Makes stars appear

4. What is wrong with the following:

/*Turns on an LED on for one second, then off for one second, repeatedly. This example code is in the public domain.*/

```
int led = 13;
void setup() {
pinMode(led, INPUT);
}
void loop() {
digitalWrite(led, HIGH);
delay(1000);
digitalWrite(led, LOW);
delay(1000);
}
```

A. All the code
B. pinMode function
C. High and Low
D. Low and High

5. A program written with the IDE for Arduino is called _____

A. IDE source
B. Sketch
C. Cryptography
D. Source code

6. How many PWM pins are there on the UNO board?

A. 4
B. 5
C. 6
D. 7

7. What is the operating frequency of Arduino UNO?

A. 12 MHz
B. 14 MHz
C. 16 MHz
D. 20 MHz

8. What is the total memory available that can be programmed in Arduino UNO Boards?

A. 16 KB
B. 31.5 KB
C. 32 KB
D. 31 KB

9. PWM stands for:

A. Pulse Width Memory
B. Pulse Width Modulation
C. Pitch Width Modulation
D. Ping With Memory

10. Arduino UNO can only be used to interface 14 LEDs.

A. True
B. False

11. How many analog pins are there in Arduino UNO?

A. 4
B. 5
C. 6
D. 7

12. delayMicroseconds() function is used to provide delay in:

A. Seconds
B. Milliseconds
C. Microseconds
D. None of above

13. "||" Stands for:

A. AND
B. OR
C. NAND
D. NOR

14. What is internet?

A. Interconnection of local area networks
B. A vast collection of different networks
C. A single network
D. None of the mentioned

15. What does UART stand for?

A. Universal Asynchronous Receiver Transmitter
B. Unique Asynchronous Receiver Transmitter
C. Universal Address Receiver Transmitter
D. Unique Address Receiver Transmitter

16. What rate can define the timing in the UART?

A. Bit rate
B. Baud rate
C. Speed rate
D. Voltage rate

17. How many Analog Inputs does Arduino Uno R3 has?

A. 2

B. 3
C. 4
D. 6

18. In the RGB LED, the R stands for _____.

A. Red
B. Read
C. Reader
D. Ready

19. The command which is used to display the characters in the serial monitor is:

A. Serial
B. Serial.type
C. Serial.println
D. Serial.printing

20. Which of the following is true about Arduino?

A. Arduino boards are able to read analog or digital input signals from different sensors.
B. Arduino IDE uses a simplified version of C++, making it easier to learn to program.
C. Arduino provides a standard form factor that breaks the functions of the micro-controller into a more accessible package.
D. All of the above.

21. Which of the following is true about IoT?

A. The term Things in the Internet of Things refers to anything and everything in day to day life.
B. IoT has greater transparency, control, and performance.
C. Both A and B.
D. None of the above.

22. What is true about memory unit?
A. A memory unit is the collection of storage units.
B. The memory unit stores the binary information in the form of bits.
C. Both A and B
D. None of the above

23. In how many categories memory/storage is classified?
A. 2
B. 3
C. 4
D. 5

24. Which of the following is correct example for Secondary Memory?
A. Magnetic disks
B. Tapes
C. EEPROM.
D. Both A and B

25. How many pins are available for interrupt functions in the Arduino Uno?
A. No pins
B. All pins except pin 2 and 3
C. Only pin 2

D. Only pins 2 and 3

26. What is the use of the detachInterrupt() function?

A. To turn off interrupt
B. To turn on interrupt
C. To disable interrupt functions for the remaining ON time.
D. To turn off the Arduino

27. What is the output of "pin1" if "pin2" is sent "1011" where 1 is 5V and 0 is 0V?

```
int pin1 = 12;
int pin2 = 11;
void setup() {
  pinMode(pin1, OUTPUT);
  pinMode(pin2, INPUT);
  Serial.begin(9600);
}
void loop() {
  if(digitalRead(pin2)==1) {
    digitalWrite(pin1,LOW);
  }
else if(digitalRead(pin2)==0) {
    digitalWrite(pin1,HIGH);
  }
}
```

A. 0100
B. 1011
C. 1110
D. 1111

28. What are the two modes that the pinMode() method sets for a particular pin?

A. DIGITAL and ANALOG
B. INPUT and OUTPUT
C. TX and RX
D. READ and WRITE

29. What are the voltage levels that can be detected using analogRead() method in the Arduino Uno?

A. 0 and 5V
B. 0 to 5.1V
C. 0 to 5V
D. 0 to 10V

30. What will the code given below give as output if a 5V line is connected as input to pin A0?

```
int pin_1=A0;
void setup() {
  pinMode(pin_1, INPUT);
  Serial.begin(9600);
}
void loop() {
  int reading=analogRead(pin_1);
  Serial.println(reading);
}
```

A. 0
B. 102
C. Null
D. 1023

Section B

Note: Attempt all questions. 40 x 0.5 = 20 Marks

31. How many errors are present in the code given below?

```
int pin1=A0;
void setup() {
  pinmode(pin1, IN);
  Serial.begin(9600);
}
void loop() {
  int value=analogRead(pin1);
  Serial.println(value+10);
}
```

A. 1
B. 2
C. 3
D. 4

32. Can the setup() function change the value of constant variables?
A. Yes, it can change
B. No, it cannot change
C. Yes, it can change but only integer values
D. Yes, it can change but only byte values

33. What are the errors in the code given below?

```
void Setup() {
  pinMode('A0',OUTPUT);
  analogwrite(100,'A0');
}
void loop() {
  //Do Nothing
}
```

A. 1
B. 2
C. 3

D. 4

34. How many voltage levels are present when a pin uses digitalWrite()?

A. 1
B. 2
C. 3
D. 4

35. Can the digitalRead() function identify analog voltages?

A. Yes, it can
B. No, it cannot
C. Yes, it can but only 2
D. Yes, it can but only 4

36. How many errors are present in the code given below?

```
int i=10;
void setup() {
    pinMode(i,OUTPUT);
    digitalwrite(HIGH);
}
void loop() {
    //Do Nothing.
}
```

A. 1
B. 2
C. 3
D. 4

37. What type of signal does the analogWrite() function output?

A. Pulse Width Modulated Signal
B. Pulse Code Modulated Signal

C. Pulse Amplitude Modulated Signal
D. Frequency Modulated Signal

38. What is the resolution of analog readings on the Arduino Uno?
A. 5V/1023
B. 3.3V/1024
C. 5V/1024
D. 3.3V/100

39. What is the maximum number of arguments that the pulseIn() function accepts?
A. 1
B. 2
C. 3
D. 4

40. What is the return type of the pulseIn() function?
A. null
B. signed byte
C. unsigned long
D. unsigned int

41. Difference between the = and the == operators?
A. Assignment & Comparison
B. Comparison & Assignment
C. Nor A neither B
D. Both A & B

42. Communication in UART is _____
A. Only simple
B. Only duplex

C. Only full duplex
D. Simplex, half duplex, full duplex

43. The Atmega328 is an _____ bit chip.
A. 8
B. 16
C. 32
D. 64

44. What is the unit of delay in the code given below?

```
void setup() {
  Serial.begin(9600);
}
void loop() {
  digitalWrite(10,HIGH);
  delay(1000);
  digitalWrite(10,LOW);
  delay(1000);
}
```

A. Milliseconds
B. Microseconds
C. Seconds
D. Minutes

45. What is the final value of the d2 variable?

```
const int trigPin = 9;
const int echoPin = 10;
long d1;
int d2;
void setup() {
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  Serial.begin(9600);
```

```
}
void loop() {
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);
  d1 = pulseIn(echoPin, HIGH);
  d2 = d1*0.034/2;
  Serial.println(d2);
}
```

A. Distance
B. Duration
C. Time Period
D. Position

46. What kind of waves does the Ultrasonic Sensor work on?
A. Gas
B. Heat
C. Light
D. Sound

47. What is the maximum distance that the Ultrasonic Sensor can detect an object?
A. 8 cm
B. 400 cm
C. 9 cm
D. 20.21 cm

48. What is the full form of SRAM?
A. Synchronized Random Access Memory
B. Static Random Access Memory

C. Simplified Random Access Memory
D. Segregated Static Random Access Memory

49. What is the use of the RESET button on the Arduino UNO?
A. To restart the code present in the Arduino
B. To restore a previous version of a code
C. To erase the code present in the Arduino
D. To erase the SRAM in Arduino Microcontroller

50. What is the use of the Vin pin present on some Arduino Boards?
A. To provide a 5V output
B. Is used for plugging in 3V supply
C. To power the Arduino Board
D. To ground the Arduino Board

51. What does the analogRead() function do physically when invoked in a code?
A. It accepts a digital Input
B. Reads the raw voltage value
C. Activates the ADC Circuit to read voltage values
D. Activates the DAC Circuit to read voltage values

52. Which function executes first when a sketch is built?
A. loop()
B. setup()
C. init()
D. main()

53. What is the use of #define?
A. To define macros
B. To define pointer variables

C. To define file names
D. To create variables

54. How many times does the loop() function run on every startup of the Arduino System?

A. 1
B. Depends upon the setup() function
C. Infinitely till the power is supplied to the Arduino
D. 3

55. How many arguments does the digitalRead() function have?

A. 1
B. 2
C. 3
D. 4

56. What is the numeric base of the math operations performed in binary?

A. 1
B. 2
C. 3
D. 4

57. What is the output of the program if the input to pin1 is 3.3V on the Arduino Due?

```
int pin1=A0;
void setup() {
  pinMode(pin1,INPUT);
  Serial.begin(9600);
}
void loop() {
```

```
  if(analogRead(pin1)==1023) {
    Serial.println("Digital HIGH Detected!");
  }
  else {
    Serial.println("Digital LOW Detected!");
  }
}
```

A. Digital HIGH Detected!
B. Digital LOW Detected!
C. Runtime Error
D. Null

58. The value 9600 is the speed at which the data will travel between the computer and the Arduino, also known as baud. This value must match the speed setting at the bottom right of the Serial Monitor. How to write the code?

A. Serial.begin(9600);
B. Serialln.begin(9600);
C. Serial.end(9600);
D. Serial.void(9600);

59. Why does the LED have different length legs?

A. For ease of inserting into the breadboard.
B. To indicate which side is negative and which side is positive.
C. To ensure it does not easily break.

60. How long is the LED on?

```
digitalWrite(13, HIGH);
delay(1000);
digitalWrite(13, LOW);
delay(1000);
```

A. 1000 seconds
B. 100 seconds
C. 1 second
D. 100 milliseconds

61. Which is wrong for the standard Arduino function setup()?

A. It is not necessary for the Arduino's program to run.
B. It runs once when the Arduino is energized.
C. It runs continuously when the Arduino is energized.
D. It will restart when the Arduino card is reset.

62. Which is wrong for the standard Arduino function loop()?

A. It is not necessary for the Arduino's program to run.
B. It runs once when the Arduino is energized.
C. It runs continuously when the Arduino is energized.
D. It will restart when the Arduino card is reset.

63. How many volts does the HC-SR04 ultrasonic distance sensor work with?

A. 1.2V
B. 3.3V
C. 9V
D. 5V

64. Which pin connection designation of the HC-HR04 ultrasonic sensor is given correctly?

A. VCC-TRIG-ECHO-GND
B. VCC-N/C-DATA-GND
C. VCC-IN-OUT-GND
D. VCC-LATCH-DATA-GND

65. What is the purpose of the pin named TRIG of HC-SR04 sensor?
A. Allows the ultrasonic sound wave to be sent from the sensor.
B. Provides the information that the ultrasonic sound wave is returned.
C. Allows the sensor to be fed with energy.
D. Provides the chassis connection of the sensor.

66. What is the purpose of the pin named ECHO of HC-SR04 sensor?
A. Allows the ultrasonic sound wave to be sent from the sensor.
B. Provides the information that the ultrasonic sound wave is returned.
C. Allows the sensor to be fed with energy.
D. Provides the chassis connection of the sensor.

67. Which of the following is true for the Arduino C code for the following Arduino UNO development board?

```
digitalWrite(13, HIGH);
```

A. Makes digital pin-13 to 0V.
B. Makes digital pin-13 to logic 0.
C. Makes digital pin-13 to 5V.
D. All other options are wrong.

68. Which of the following is the Arduino C command that reads the logic value in digital pin 13?
A. digitalWrite(13, 1)
B. pinMode(13, INPUT);
C. pinMode(13, OUTPUT);
D. digitalRead(13);

69. How many GND pins are available with soldered on the Arduino Uno development board?

A. 1
B. 2
C. 3
D. 4

70. Find correct library for Servo motor.

A. #include<servo.h>
B. #include<Servo_motor.h>
C. #include<Servo.h>
D. None of the above.